

SESSION 9

Sources of Finance

Angels · Accelerators · Crowdfunding

Entrepreneurial Finance & Venture Capital

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1 · Course Overview — The Two Big Questions

The first half of the course established the foundations. From here the focus shifts to **who should you take money from** — and how their incentives compare with yours.

Question	What We Covered
How much external finance does this business need?	Business model, cash curve, burn rate, maximum financing need
How should I structure my financing?	Round sizing, multi-stage financing, valuation, term sheets & deal terms
Who should I take money from? (NOW)	Incentives of each investor type, alignment / misalignment with entrepreneur

Stage	Typical Sources
Seed / Idea Stage	Friends & Family, Angels (SAFE/Conv. Note), Crowdfunding, Accelerators
Early Stage (Product-Market Fit)	Seed VC, Venture Capital (Preferred Stock), Venture Debt, Customer Financing
Later Stage & Exit	VC, Growth Equity, Venture Debt, Receivables Financing, Corporate/Strategic, IPO

KEY THEME

Each source carries a different level of dilution and different incentives. Understanding **whether those incentives are aligned or misaligned with yours** is the central question from here on.

2 · Angel Investors

Angels invest their own money — not a fund's. This single difference drives almost everything about how they behave.

Characteristic	Implication
Invest own money	Cannot follow on in later rounds — pro-rata becomes unaffordable as valuation rises
Earlier stage, smaller cheques	Take on more risk; company may have no product or revenue yet
No institutional LP pressure	More flexible on terms, but also less sophisticated in some cases
Extremely dilution-sensitive	Must target massive potential outcomes to justify the dilution they will suffer

The dilution problem in practice:

WORKED EXAMPLE

If you invest £100k at a £1M post-money (10% ownership), and the company next raises at a £100M valuation, your pro-rata is £10M — which most angels cannot provide. Your ownership shrinks, and large institutional investors often don't want small angels on the cap table anyway.

The base rates (Scott Shane, Fool's Gold):

Metric	Number
US companies per year projecting \$10M revenue in 5 years	~80,000
Number that actually achieve \$10M+ in sales	~3,500
Share of angel investments that IPO	~0.2%
Share that are acquired	~1%

In aggregate, angels generate poor returns. Sophisticated angels who understand what we cover in this course can make money — but it is the exception, not the rule. **If you anticipate heavy dilution ahead, you must invest only in companies with the potential for massive outcomes.**

3 · The Falling Cost of Experiments — Why It Changed Everything

The biggest shift in early-stage entrepreneurship over the last 20 years has been a dramatic fall in the cost of running the first experiment — the first round of financing — before you know whether your idea works.

The AWS shock (Ewens, Nanda & Rhodes-Kropf, JFE 2018):

Before Amazon Web Services (2006), starting an internet company required purchasing servers, databases, and networking gear — easily \$5–10M just to get going. After AWS, startups could rent compute at 100x–1000x lower per-unit cost, converting large fixed capital expenditures into flexible variable costs. This did *not* lower total operating costs at scale — it lowered the cost of **initial experiments**, before you know if the idea works.

The same shift is happening across sectors:

Technology Shift	Sector Benefiting
Cloud computing (AWS)	SaaS / Internet businesses
Rapid prototyping (3D printing)	Hardware startups
Gene sequencing, CRISPR, AlphaFold	Biotechnology
Supercomputing / simulation	Nuclear power (new reactor designs)
Reusable rockets (SpaceX)	Space tech / CubeSats
AI / large language models	Almost everything — reducing cost of building & staffing

PROF. NANDA EXAMPLES FROM CLASS

A former student launched two CubeSats for his space startup having raised less than \$5M. This was impossible when it cost hundreds of millions to build a large satellite. Nuclear power: simulation reduced the cost of testing new reactor designs from several billion to several hundred million. Same logic, every sector.

What does AI change? (The Hidden Figures analogy)

AI AND THE HIDDEN FIGURES ANALOGY (FROM LECTURE)

In the film *Hidden Figures*, a department of human 'computers' at NASA (1960s) watched IBM machines arrive and realised their jobs were threatened. But the same people went on to programme those machines — becoming complements to the technology rather than substitutes. The pattern repeats every technological shift: whatever the technology substitutes for disappears; what is *complementary* to it flourishes.

AI compresses the cost of starting businesses further — websites, code, marketing, and research can all be done with far fewer people. AI also reduces the cost of scientific experiments, digitally searching vast molecular or material possibility spaces before committing to physical tests.

Key distinction in the current moment: The **model companies** (OpenAI, Anthropic, etc.) are extremely capital intensive — they are the AWS of this cycle. The **application-layer startups** building on top of those models are the cheap-experiment beneficiaries, just as web startups benefited from cloud computing. These are very different businesses with very different capital structures.

4 · Multi-Stage Financing: The Real Options Framework

Each funding round is an **experiment** that generates information. If intermediate results are positive, you invest more. If negative, you abandon. This is a real option — and when the cost of the first experiment falls, the value of that option rises.

The minimum viable probability — key formula:

A VC invests \$X in Stage 1 only if the expected payoff exceeds the investment:

INVESTMENT CONDITION

$$p1 \times (pS \times V - Y) - X > 0$$

MINIMUM PROBABILITY OF STAGE 1 SUCCESS

$$p1_min = X / (pS \times V - Y)$$

Where: **p1** = probability of positive intermediate result, **pS** = probability of ultimate success given positive result, **V** = exit value, **Y** = Stage 2 investment, **X** = Stage 1 investment.

Class worked example — base case (X = \$5M):

Stage 2: 40% chance of \$300M exit, VC2 invests \$20M. Stage 1: 10% chance of positive intermediate results, VC1 invests \$5M.

	Stage 2 (VC2)	Stage 1 (VC1)	Entrepreneur
Post-money valuation	\$120M (= 40% x \$300M)	\$10M (= 10% x \$100M)	—
Investment	\$20M	\$5M	—
Ownership at Stage	16.7% (= 20/120)	50% (= 5/10)	50%
Ownership at exit (after dilution)	16.7%	41.7% (= 50% x 83.3%)	41.7%

What if X falls from \$5M to \$2M?

Stage 2 is unchanged (VC2 still takes 16.7%). The post-money at Stage 1 is still \$10M — but now VC1 only needs $\$2/\$10 = 20\%$. Pre-money rises to \$8M. The entrepreneur captures the surplus, ending up with ~66% rather than 41.7%.

Scenario	Stage 1 Cost	VC1 Ownership	Entrepreneur at Exit	Pre-money S1
Base case	\$5M	50%	41.7%	\$5M
Cost falls	\$2M	20%	~66%	\$8M
VC keeps same IRR	\$2M	50%	41.7%	\$4M (min prob. falls to 4%)

KEY INTUITION

The **step-up in value** is inversely related to the probability of Stage 1 success. A company with only a 4% chance of getting to Stage 2 will generate a *much larger* step-up if it does succeed — because investors update their beliefs more strongly when an unlikely thing happens.

5 · Three Consequences of Falling Experiment Costs

1

Wider funnel — more companies get funded

When $p1_min$ falls, companies with much lower probabilities of early success become NPV-positive investments. These are 'long shot bets' — low probability of a huge outcome. Before AWS, a VC might only fund companies with a 10% Stage 1 success probability. After AWS, they can fund companies with a 4% probability and still hit the same IRR threshold.

2

More value retained by entrepreneurs

In competitive VC markets, falling experiment costs get bid up as higher pre-money valuations. The entrepreneur captures the surplus. In the worked example, the founder's exit ownership rises from 41.7% to ~66% as Stage 1 cost falls from \$5M to \$2M.

3

Fewer gatekeepers

If the first round only needs \$500K instead of \$5M, founders don't need large institutional VCs. Angels, micro-VCs, and friends & family can provide it. This is particularly valuable for entrepreneurs outside established networks (outside Silicon Valley, or without connections to major VC firms) — they now have a realistic shot at early funding.

6 · Empirical Evidence — Software vs. Biotech

Data across US VC-backed companies (Nanda 2024) shows a striking contrast between sectors with cheap vs. expensive early experiments. $\text{Step-up} = \text{Pre-money}(t+1) / \text{Post-money}(t)$.

Characteristic	Software	Biotech / Pharma
Typical Round 1 deal size	~\$2M	~\$5–10M
Round 1 step-up (avg)	>2.0x	~1.65x

Round 5 deal size	~\$15–20M	~\$35–40M
Round 5 step-up (avg)	~1.4x	~1.2x
Early experiment reveals...	A lot — market demand, usage, retention	Little — Phase I safety data only
NPV-positive investment universe	Very large (cheap experiments)	Small (expensive experiments)

Why invest in biotech at all?

WHY BIOTECH IS DIFFERENT

If experiments are expensive and step-ups are low, the **exit value V must be enormous** to compensate. US patent protection + mandatory insurance reimbursement creates multi-billion dollar blockbuster drugs. Without that, the math simply doesn't work. This is also why diseases without a paying patient population (malaria, many tropical diseases) are chronically underfunded — the V is too low relative to the investment required.

INTERACTION WITH PAY-TO-PLAY

Pay-to-play provisions make biotech investing even harder for angels and early investors who cannot follow on across multiple expensive rounds. This raises the IRR threshold further and makes the required exit value even larger.

7 · VC Response: 'Spray and Pray' Investing (Ewens, Nanda & Rhodes-Kropf, JFE 2018)

Using a difference-in-differences design (AWS-affected sectors vs. unaffected sectors, before and after 2006), this paper documents how the falling cost of experimentation changed VC behaviour:

Outcome Variable	Finding
Initial investment size	15–27% smaller in affected sectors after 2006
Number of first investments per VC	Nearly doubled in affected sectors
Follow-on rate (round 2)	Fell — VCs more willing to abandon after round 1
Board seat participation (round 1)	14–21% lower probability of board seat
Founder age / experience	Fell — more young, first-time founders funded
Failure rate	Higher in affected sectors post-2006
Step-up in value (for survivors)	15–20% larger — consistent with long shot bets

INTERPRETATION

This is the 'spray and pray' investment strategy. Instead of deep governance and high investment in a few companies, VCs make many small bets and let the experiment decide. This makes sense when experiments are cheap: you learn a lot from a little. The step-up result is key — those that succeed generate *much* larger valuation jumps, consistent with long shot bets not worse companies.

Side effect — the governance gap:

Spray and pray means VCs take fewer board seats and provide less governance at precisely the stage when inexperienced founders need it most. This created demand for **accelerators** — scalable mentorship intermediaries that fill the governance gap.

8 · Accelerators

Accelerators emerged as a direct response to the changing needs of early-stage founders. Y Combinator (founded 2005) was the first; today there are hundreds worldwide.

Feature	Detail
Structure	Cohort-based bootcamp, typically 3 months, culminating in Demo Day
Focus	Customer discovery, product-market fit, investor pitch preparation
Economics	Take 7–10% equity in exchange for cash (\$25K–\$100K) + network access
Share class	Common stock (not preferred) — avoids awkward comparison to Series A pricing
Additional capital	Some add convertible note / SAFE on top of equity

Why common stock matters:

The accelerator typically gets in at a tiny post-money valuation (e.g. \$333K for Airbnb's YC entry). Three months later the company raises at \$3M. That's a 7x step-up. VCs could argue this is suspicious — but because YC took **common stock** and Series A investors take **preferred stock**, they are different share classes and not directly comparable. This is the practical reason why accelerators almost always take common.

Who benefits most from accelerators?

The trade is giving away 7–10% equity in exchange for mentorship, network, and credibility. This is only worth it if what you receive more than compensates:

Founder Type	Is Accelerator Worth It?
Technical founder, no business/investor network	Yes — the network and credibility are worth 7–10%
First-time entrepreneur who doesn't 'speak VC'	Yes — learning the language and getting intro'd is valuable
Serial entrepreneur with established networks	Probably not — giving up 10% for introductions you already have
Outside Silicon Valley / without connections	Especially valuable — opens doors that would otherwise be closed

9 · Case Study: Airbnb Financing Rounds

Airbnb is the canonical illustration of how accelerators add value, how step-ups work across rounds, and how the combination of a great team and patient capital creates enormous value. The Paul Graham / Fred Wilson email exchange (Jan–Feb 2009) is a real-world lesson in the difficulty of identifying long shot bets *ex ante*.

Round	Date	Investment (\$K)	% Taken	Post-Money (\$K)	Step-Up
YC Seed	Jan 2009	\$20	6.0%	\$333	—
Seed (Sequoia)	Apr 2009	\$600	20.0%	\$3,000	7.2x
Series A	Nov 2010	\$7,200	7.2%	\$100,000	31x
Series B	Jun 2011	\$117,052	9.0%	\$1,300,000	11.8x
Series C	Oct 2012	\$200,000	8.0%	\$2,500,000	1.8x
Series D	Apr 2014	\$475,000	4.8%	\$10,000,000	3.8x
Series E	Jun 2015	\$1,500,000	5.9%	\$25,500,000	2.4x
Series F	Sep 2016	\$555,500	1.9%	\$29,700,000	1.2x

YC's return:

KEY LESSON

YC invested \$20K for 6% equity — worth \$333K post-money. At the Series F (\$29.7B valuation), YC's 3% (after dilution) was worth approximately **\$877M**. The reason: Airbnb raised at extremely high step-ups in its early rounds, so YC was diluted much less than a company that raises flat or at modest step-ups. This is the compound effect of high step-ups — the early investor retains far more value.

THE FRED WILSON STORY

Fred Wilson at Union Square Ventures passed on investing in Airbnb despite Paul Graham's strong recommendation in early 2009. He couldn't get his head around 'air mattresses on living room floors' as the next hotel room. This is the classic long shot bet — genuinely hard to see ex ante, with a huge payoff for those who could. The founders retained ~51% through eight rounds of funding totalling \$2.9B — because every round was priced at a significant step-up.

10 · Crowdfunding

Crowdfunding has existed for centuries in donation form (churches, charities). The modern innovation is bringing it online with new structures.

Type	How It Works	Key Feature
Donation	Give money with no expectation of return	Has existed forever; good for causes
Reward-based (Kickstarter)	Back a product idea; get the product when made	Signals demand AND provides customer financing (negative asset intensity)
Lending (peer-to-peer)	Lend to small businesses at interest	Upside is capped — you get interest, nothing more
Equity crowdfunding	Buy shares in private companies	New; contentious — see below

The reward-based model (Kickstarter) — two benefits:

1. **Signal of demand** — before spending capital on production, you learn whether customers actually want it.
2. **Negative asset intensity** — customers pay upfront (like SaaS subscriptions or airline tickets). The customer is financing you, reducing external capital needs. The asset intensity ratio goes negative.

Equity crowdfunding — the policy debate:

POLICY DEBATE

Historical reason for restriction: Following the Great Depression, the US government restricted private company investment to 'accredited investors' (wealthy individuals) — to protect unsophisticated investors from fraud and the complexity of these transactions.

Modern counter-argument: More companies are staying private longer. Value accrual increasingly happens pre-IPO. Restricting private market access to the wealthy *perpetuates inequality* — the rich get to participate in growth that the non-wealthy are locked out of.

The tension: These are genuinely complex, bilateral transactions with many ways to be taken advantage of. Yet restricting access concentrates wealth. Both perspectives have merit.

ONE SOLUTION

One partial solution: pension funds. If pension fund managers — acting as sophisticated institutional investors — channel retirement savings into venture capital, non-wealthy individuals get indirect exposure. This is a live policy issue in the UK (government wants more pension money in VC; pension funds are reluctant).

PROF. NANDA'S WARNING

Always ask: why me? When someone offers you an investment opportunity in a private company, ask yourself: why are they coming to me and not to other investors? Have they already been to better-connected people who said no? The private market has no market-clearing price — unlike public markets, you cannot rely on the price being correct.

11 · Key Takeaways

1

Investor incentives vary fundamentally

Angels invest own money and can't follow on — extremely dilution-sensitive. VCs invest others' money with IRR targets and fund return obligations. Each source shapes how aligned they are with your interests.

2

Multi-stage financing = real options

Each round is an experiment. The option to abandon after bad intermediate results is valuable — and gets more valuable as experiment costs fall.

3

Falling experiment costs expand the investable universe

When X (Stage 1 cost) falls, $p1_min$ falls. More risky 'long shot bets' become fundable. This is what AWS did for software, CRISPR for biotech, reusable rockets for space.

4	Entrepreneurs capture falling experiment costs as higher pre-money valuations In a competitive VC market, lower first-round costs translate to higher pre-money valuations, not just lower VC ownership. The surplus accrues to the entrepreneur.
5	Step-up and experiment cost are inversely related Companies that succeed when they had low odds of success generate larger valuation jumps. This is why software has high step-ups and biotech has low ones.
6	Accelerators fill the governance gap created by spray and pray They emerged because VCs were writing smaller cheques with less governance — precisely when inexperienced founders needed mentorship most. Worth it for technical founders without networks; not for serial entrepreneurs who have them.
7	Reward-based crowdfunding reduces capital needs Customer pre-payment before production creates negative asset intensity. Less external capital needed, and you get a demand signal for free.
8	Private market access inequality is a live policy issue Restricting equity crowdfunding to accredited investors protects unsophisticated investors but concentrates wealth creation among the wealthy. Pension fund channelling is one partial fix.